Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems

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For any N-fermion system whose single-particle orbitals are products of spherical harmonics and arbitrary radial functions, the fraction of nonzero two-body matrix elements is determined solely by angular momentum selection rules and is independent of the radial potential. Exact enumeration of the Gaunt coefficients yields the angular ERI density $D(l_{\max})$ under the Coulomb selection rule (conservation of total magnetic quantum number, $m_a + m_b = m_c + m_d$): 14.84% at $l_{\max} = 1$, 8.52% at $l_{\max} = 2$, 6.06% at $l_{\max} = 3$, 4.83% at $l_{\max} = 4$, and 3.99% at $l_{\max} = 5$. At $l_{\max} = 2$, 91.48% of all possible two-body matrix elements are guaranteed to vanish by the Wigner–Eckart theorem before any property of the potential is invoked. A stricter pair-diagonal restriction ($m_a = m_c$, $m_b = m_d$), applicable to axially-symmetric or m-decoupled bases, gives a further reduction (2.76% at $l_{\max} = 2$, 1.44% at $l_{\max} = 3$). The bound is independent of the number of radial quantum numbers retained at fixed $l_{\max}$, and therefore independent of qubit count. Numerical verification across five potentials (Coulomb, harmonic oscillator, Woods–Saxon, square well, Yukawa) confirms the theorem: angular sparsity patterns are bit-identical across all potentials at matched orbital counts. The radial-angular factorization and the Gaunt selection rules are standard results from Condon & Shortley and Slater; the contribution of this paper is the explicit quantitative bound at each $l_{\max}$, framed as a universal resource guarantee for qubit Hamiltonian sparsity that applies to electronic structure, nuclear shell models, quantum dots, and any other spherical-potential fermion system.

I. INTRODUCTION

Quantum simulation of fermionic systems requires encoding second-quantized Hamiltonians as sums of Pauli operators. The number of Pauli terms, the number of qubitwise-commuting (QWC) measurement groups, and the Pauli 1-norm λ are the dominant resource metrics for near-term variational [1] and fault-tolerant [2] quantum algorithms. All three metrics scale with the number of nonzero two-body matrix elements in the chosen orbital basis, making Hamiltonian sparsity the central structural resource for quantum simulation.

Existing sparsity analyses are potential-specific. In quantum chemistry, the Coulomb $1/r_{12}$ interaction produces sparsity governed by the multipole expansion and the basis set's spatial locality [3, 4]. In nuclear structure physics, the nucleon–nucleon (NN) interaction in a harmonic oscillator (HO) basis produces sparsity governed by Talmi–Moshinsky brackets and center-of-mass factorization [5]. These analyses are carried out independently, obscuring a structural feature that is common to both domains: the angular momentum selection rules imposed by the spherical harmonic component of the basis are identical.

In this paper we state the resulting potential-independent sparsity bound explicitly. For any rotationally invariant (central) two-body interaction expanded in a spherical harmonic basis, the zero structure of the two-body matrix elements depends only on the angular quantum numbers (l, m) and is independent of the radial wavefunctions, the radial potential, and the radial quantum numbers. The underlying factorization of two-body integrals into angular (Gaunt) coefficients and radial (Slater-type) integrals has been standard since Condon and Shortley [6] and Slater [7], and we claim no mathematical originality for it. The contribution of this paper is the explicit quantitative ERI density at each $l_{\max}$, verified numerically across five potentials, and framed as a universal resource bound for the quantum computing community.

This result was identified in the course of extending the GeoVac spectral graph theory framework [9, 10] from electronic to nuclear structure applications. In the companion paper [13] we show that a separate feature of the GeoVac framework—namely the convergence of the one-body graph Laplacian spectrum to the pure integers of the continuum S^3 Laplace–Beltrami operator—is special to the Coulomb potential, so that the universal–specific partition identified here is sharp: angular sparsity is universal across spherical fermion systems, while the S^3 Fock projection and the π-free graph structure are Coulomb-specific.

II. TWO-BODY MATRIX ELEMENT FACTORIZATION

Consider N fermions in a single-particle basis $\{|n, l, m\rangle\}$ where n labels the radial quantum number, l is the orbital angular momentum, and m is the magnetic quantum number. The spatial orbitals are

$$\phi_{nlm}(\mathbf{r}) = R_{nl}(r)\, Y_l^m(\hat{r}), \tag{1}$$

where $R_{nl}(r)$ are radial wavefunctions determined by the single-particle potential $V(r)$ and Y_l^m are spherical harmonics.

Let $\hat{V}_{12} = V(|\mathbf{r}_1 - \mathbf{r}_2|)$ be a rotationally invariant two-body interaction. We expand $\hat{V}_{12}$ in Legendre polynomials:

$$V(|\mathbf{r}_1 - \mathbf{r}_2|) = \sum_{k=0}^{\infty} v_k(r_1, r_2)\, P_k(\cos\theta_{12}), \tag{2}$$

where $v_k(r_1, r_2)$ are radial expansion coefficients that depend on the specific form of V. For the Coulomb interaction, $v_k = r_<^k / r_>^{k+1}$; for other potentials, v_k takes different forms but the angular structure is identical.

Theorem 1 (Radial–Angular Factorization). *For any rotationally invariant two-body interaction $V(|\mathbf{r}_1 - \mathbf{r}_2|)$, the two-body matrix element factorizes as*

$$\langle n_1 l_1 m_1,\, n_2 l_2 m_2 \,|\, \hat{V}_{12} \,|\, n_3 l_3 m_3,\, n_4 l_4 m_4 \rangle = \sum_k c^k(l_1 m_1;\, l_3 m_3)\, c^k(l_2 m_2;\, l_4 m_4)\, R^k(n_1 l_1, n_2 l_2;\, n_3 l_3, n_4 l_4), \tag{3}$$

where the Gaunt coefficients

$$c^k(l\,m;\, l'\,m') = (-1)^m \sqrt{(2l+1)(2l'+1)} \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l' \\ -m & q & m' \end{pmatrix} \tag{4}$$

with $q = m - m'$ depend only on angular quantum numbers, and the Slater-type radial integrals

$$R^k = \int_0^\infty \int_0^\infty R_{n_1 l_1}(r_1)\, R_{n_3 l_3}(r_1)\, v_k(r_1, r_2)\, R_{n_2 l_2}(r_2)\, R_{n_4 l_4}(r_2)\, r_1^2 r_2^2\, dr_1\, dr_2 \tag{5}$$

depend on the potential V through the radial wavefunctions and the expansion coefficients v_k. The sum over k is bounded: $\max(|l_1 - l_3|, |l_2 - l_4|) \le k \le \min(l_1 + l_3, l_2 + l_4)$, *with the parity constraints $l_1 + l_3 + k$ even and $l_2 + l_4 + k$ even.*

Proof. Substituting the Legendre expansion (2) into the two-body integral and applying the spherical harmonic addition theorem $P_k(\cos\theta_{12}) = \frac{4\pi}{2k+1} \sum_q Y_k^{q*}(\hat{r}_1) Y_k^q(\hat{r}_2)$ yields a product of two angular integrals (Gaunt integrals) and one radial integral for each k. The Gaunt integral $\int Y_{l_1}^{m_1 *} Y_k^q Y_{l_3}^{m_3}\, d\Omega$ evaluates to a product of Wigner $3j$ symbols by the Wigner–Eckart theorem [8], giving Eq. (4). The bounds on k follow from the triangle inequality of the $3j$ symbol: $|l - l'| \le k \le l + l'$, applied to both Gaunt coefficients. The parity constraint follows from the $3j$ symbol $(l\ k\ l';\ 0\ 0\ 0)$, which vanishes unless $l + k + l'$ is even [6]. □

The factorization (3) is a standard result of angular momentum algebra [6–8]. The key observation for quantum computing applications is the following:

Theorem 2 (Potential-Independent Sparsity). *A two-body matrix element $\langle n_1 l_1 m_1, n_2 l_2 m_2 | \hat{V}_{12} | n_3 l_3 m_3, n_4 l_4 m_4 \rangle$ vanishes identically for every central potential whenever $c^k(l_1 m_1; l_3 m_3) \cdot c^k(l_2 m_2; l_4 m_4) = 0$ for every allowed k. Conversely, whenever at least one Gaunt product is nonzero, the matrix element is generically nonzero. The zero condition depends only on $(l_1, m_1, l_2, m_2, l_3, m_3, l_4, m_4)$ and is independent of the radial potential $V(r)$, the radial wavefunctions $R_{nl}(r)$, and the radial quantum numbers (n_1, n_2, n_3, n_4).*

Proof. The symmetry direction is immediate: if all angular factors vanish in Eq. (3), every term in the sum is zero regardless of R^k. This is the statement that the zero pattern generated by angular selection rules is a lower bound on sparsity.

For the converse, suppose there exists at least one k^* for which $c^{k^*}(l_1 m_1; l_3 m_3) \cdot c^{k^*}(l_2 m_2; l_4 m_4) \neq 0$. The corresponding Slater integral R^{k^*} is a weighted inner product of the radial functions $R_{n_1 l_1} R_{n_3 l_3}$ and $R_{n_2 l_2} R_{n_4 l_4}$ mediated by $v_{k^*}(r_1, r_2) > 0$ on a set of positive measure (true for any physical central potential that does not vanish identically in its multipole component). For any Sturm–Liouville potential with a discrete spectrum—including Coulomb, harmonic oscillator, Woods–Saxon, square well, and Yukawa—the radial products $R_{nl}(r) R_{n'l'}(r)$ are linearly independent as a function of r for distinct index tuples, and the bilinear functional defined by v_{k^*} is nondegenerate on this family. Therefore $R^{k^*} = 0$ can hold only for a measure-zero subset of the space of central potentials—pathological choices in which two or more R^k contributions conspire to cancel accidentally. Any infinitesimal perturbation of the potential restores $R^{k^*} \neq 0$. □

Remark. The theorem classifies zero matrix elements into two types: *symmetry-protected zeros* (those forced by the Gaunt selection rules) and *accidental zeros* (those produced by fine-tuned cancellations in R^k for specific potentials). Only symmetry-protected zeros transfer across potentials; accidental zeros are fragile under perturbation. Below, when we refer to the "angular ERI density" $D(l_{\max})$, we mean the density after symmetry-protected zeros are removed.

III. ERI DENSITY BOUNDS

Definition. The *angular ERI density* $D(l_{\max})$ is the fraction of four-index combinations $(l_1 m_1, l_2 m_2, l_3 m_3, l_4 m_4)$ with $l_i \le l_{\max}$ for which the matrix element is not forced to vanish by Gaunt selection rules. Equivalently, $D(l_{\max})$ is the fraction of four-index combinations for which (i) m-conservation

$m_1 + m_2 = m_3 + m_4$ holds, and (ii) at least one value of k satisfies the triangle and parity conditions for both Gaunt factors with both factors nonzero.

The total number of angular orbital labels at $l_{\max}$ is $N_{\text{orb}} = (l_{\max} + 1)^2$, so the total number of unrestricted four-index combinations is N_{orb}^4. The angular ERI density counts the fraction that survives Gaunt selection rules.

Theorem 3 (ERI Density at Low $l_{\max}$). *Exact enumeration of the Gaunt selection rules gives:*

$l_{\max}$	Shells	N_{orb}	N_{orb}^4	$D(l_{\max})$	$D_{\text{pd}}(l_{\max})$
0	s	1	1	100.00%	100.00%
1	$s+p$	4	256	14.8438%	7.8125%
2	$s+p+d$	9	6,561	8.5200%	2.7587%
3	$s+p+d+f$	16	65,536	6.0608%	1.4404%
4	$s+p+d+f+g$	25	390,625	4.8274%	0.9024%
5	through h	36	1,679,616	3.9863%	0.6190%

$D(l_{\max})$ is the physical Coulomb density (conservation of total magnetic quantum number $M_L = m_a + m_b = m_c + m_d$); D_{pd} is the stricter pair-diagonal restriction ($m_a = m_c$, $m_b = m_d$ per pair) that holds only for axially-symmetric or m-decoupled bases. These values are obtained by exact enumeration of the N_{orb}^4 four-index combinations, applying the conditions of the definition above. Both conventions are verified by exact enumeration in `tests/test_paper22_density.py`: the global-M_L D via the production Gaunt routine `casimir_ci._gaunt_ck`, and the pair-diagonal D_{pd} realized by `angular_zero_count` in `geovac/nuclear/potential_sparsity.py`.

The density decreases monotonically with $l_{\max}$: the number of orbitals grows as $(l_{\max} + 1)^2$ and the number of four-index combinations as $(l_{\max} + 1)^8$, while the number of Gaunt-allowed transitions grows more slowly (polynomially). The asymptotic behavior $D(l_{\max}) \to 0$ as $l_{\max} \to \infty$ reflects the fact that most high-l four-index combinations fail the triangle and parity conditions for every k.

Independence from radial quantum numbers. When the full orbital set includes $n_{\max}$ radial quantum numbers for each (l, m), the total number of spatial orbitals becomes $N_{\text{spatial}} = n_{\max} \times (l_{\max} + 1)^2$. The total ERI count scales as $N_{\text{spatial}}^4 = n_{\max}^4 \times N_{\text{orb}}^4$. By Theorem 2, every angular-forbidden element remains zero for all values of the radial quantum numbers, and every angular-allowed element is generically nonzero. The ERI density at fixed $l_{\max}$ is therefore independent of $n_{\max}$:

$$D(l_{\max}, n_{\max}) = D(l_{\max}) \quad \forall\, n_{\max} \geq 1. \tag{6}$$

This means the angular sparsity guarantee is independent of the qubit count $Q = 2 \times n_{\max} \times (l_{\max} + 1)^2$ at fixed $l_{\max}$.

IV. NUMERICAL VERIFICATION

To verify Theorems 1–3, we constructed single-particle bases and computed all two-body matrix elements for five distinct radial potentials at $n_{\max} = 2$, $l_{\max} = 2$ ($N_{\text{spatial}} = 18$ orbitals, $Q = 36$ qubits):

1. **Coulomb:** $V(r) = -Z/r$ (hydrogen-like, electronic structure)
2. **Harmonic oscillator:** $V(r) = \frac{1}{2}\omega^2 r^2$ (nuclear shell model, quantum dots)
3. **Woods–Saxon:** $V(r) = -V_0/(1 + e^{(r-R_0)/a})$ (nuclear mean field)
4. **Square well:** $V(r) = -V_0\,\Theta(R_0 - r)$ (textbook benchmark)
5. **Yukawa:** $V(r) = -V_0\, e^{-\mu r}/r$ (screened Coulomb, meson exchange)

For each potential, we solved the radial Schrödinger equation numerically (or analytically where available) to obtain radial wavefunctions $R_{nl}(r)$, then computed all $N_{\text{spatial}}^4 = 104{,}976$ two-body matrix elements using the Coulomb interaction $1/|\mathbf{r}_1 - \mathbf{r}_2|$ as the two-body operator. (The single-particle potential determines the *basis*; the two-body operator is the standard Coulomb repulsion in all cases. This is the physically relevant scenario for electronic structure. For nuclear structure, the two-body NN force is itself expanded in Legendre polynomials, and the same factorization applies.)

The results (Table I) confirm Theorem 2 to machine precision:

- The angular zero pattern is *bit-identical* across all five potentials. Not merely close—every element that is zero for Coulomb is zero for all other potentials, and every nonzero element is nonzero for all potentials.

TABLE I. ERI sparsity decomposition at $n_{\max} = 2$, $l_{\max} = 2$ ($N_{\text{spatial}} = 18$). Angular zeros are determined by Gaunt selection rules; radial zeros count additional vanishings from radial integral structure. The ERI-density column is the pair-diagonal D_{pd} (the density the composed pipeline realizes); the global-M_L Coulomb density D is $\sim 3\times$ larger (Theorem 3).

Potential	Angular zeros (%)	Radial zeros (%)	ERI density (%)
Coulomb	97.24	0.00	2.76
Harmonic osc.	97.24	0.00	2.76
Woods–Saxon	97.24	0.00	2.76
Square well	97.24	0.00	2.76
Yukawa	97.24	0.00	2.76

- Radial zeros contribute exactly 0.00% additional sparsity. No Slater integral R^k accidentally vanishes for any of the five potentials tested.

- The ERI density is 2.76% for all potentials, matching the pair-diagonal $D_{\rm pd}$ column of Theorem 3 at $l_{\max} = 2$ (the global-M_L density there is $D = 8.52\%$).

The bit-identical angular patterns are not a numerical coincidence: they are a direct consequence of the Gaunt coefficients being pure functions of (l, m) quantum numbers, evaluated identically regardless of what potential produced the radial wavefunctions.

V. IMPLICATIONS FOR QUANTUM SIMULATION

A. Universal resource bound

Theorem 2 provides a *potential-independent* upper bound on the number of nonzero two-body Pauli terms in any qubit Hamiltonian constructed from a spherical harmonic basis at angular momentum cutoff $l_{\max}$. At $l_{\max} = 2$, at most 8.52% of all possible two-body terms are nonzero; at $l_{\max} = 3$, at most 6.06%. This bound applies identically to:

- *Electronic structure:* Coulomb repulsion in hydrogen-like or Slater-type bases [10].

- *Nuclear shell model:* NN interaction in the harmonic oscillator *j*-*j* basis [5]. The central component of the NN force respects the same Gaunt selection rules.

- *Quantum dots:* Electron–electron repulsion in parabolic confinement bases [14].

- *Trapped ions and cold atoms:* Any system where single-particle states are expanded in spherical harmonics.

The bound means that qubit Hamiltonian sparsity estimates derived for one physical domain transfer directly to others at matched $l_{\max}$, without re-analysis of the interaction potential.

B. Composed architectures: $O(Q^{2.5})$ Pauli scaling for arbitrary spherical potentials

The angular sparsity of Theorem 2 applies to single-center orbital bases. In the GeoVac composed architecture [10, 11], a molecular Hilbert space is decomposed into block-diagonal orbital subspaces (core, bond, lone pair), each centered on a specific nucleus and each carrying its own spherical harmonic basis. Cross-block electron repulsion integrals are suppressed structurally by the spatial factorization, and within-block integrals are subject to the angular selection rules of Theorem 2. The resulting Pauli scaling measured across the LiH/BeH_2/H_2O/HF/NH_3/CH_4 library is $O(Q^{2.5})$ [10], with exponent spread less than 0.05 across molecules.

Corollary 1 (Potential-Independent $O(Q^{2.5})$ Composed Sparsity). *In any composed block architecture whose per-block single-particle bases are spherical harmonic products $R_{nl}(r)Y_l^m$ and whose inter-block couplings respect the block decomposition, the Pauli scaling $O(Q^{2.5})$ observed for the Coulomb interaction in Ref. [10] reflects the set of surviving ERIs, which is potential-independent (Theorem 2): the radial potential affects only the magnitudes of the surviving ERIs, not which ERIs survive. The* 2.5 *exponent itself is the measured value for the Coulomb hydride library and is expected, but not here proven, to carry over to other central interactions.*

Proof sketch. The composed Pauli scaling decomposes into a block-diagonal factor (same for all potentials, depending only on the fiber-bundle structure) and a within-block angular factor (identical for all central potentials by Theorem 2). The radial integrals R^k merely renormalize the coefficients of surviving Pauli strings. Since the Pauli count is determined by which strings have nonzero coefficient, the count is potential-independent; the Pauli 1-norm λ and the QWC group count may differ, but the number of terms and the scaling exponent match. □

This corollary means that the angular selection-rule sparsity *pattern* of the GeoVac composed architecture—the density that yields the 51× to 1,712× fewer Pauli terms measured against published Gaussian baselines for electronic structure—is present for nuclear shell model Hamiltonians, quantum dot Hamiltonians, and any other composed block architecture built from spherical harmonic bases. The specific 51×–1,712× ratios are measured Coulomb-hydride-vs-Gaussian numbers, expected but not here proven to carry over quantitatively. The angular selection rule structure is a computational resource that belongs to the geometry of angular momentum, not to the Coulomb potential.

C. $\mathbf{Z}_2$ tapering from Gaunt parity

The Gaunt parity selection rule $(\ell_a+\ell_c+k)$ even AND $(\ell_b+\ell_d+k)$ even in Theorem 2 forces $\ell_a+\ell_b+\ell_c+\ell_d$ to be even on every nonzero ERI element. Therefore the operator $P_\ell^{(i)} = (-1)^{N^{(i)}_{\ell\text{-odd}}}$, counting electrons in ℓ-odd orbitals of sub-block i, commutes with the two-body Coulomb sector exactly, providing a $\mathbb{Z}_2$ qubit-tapering stabiliser independent of the Hopf $m_\ell \to -m_\ell$ reflection. Per-sub-block, this saves $\Delta Q = n_{\rm sb}$ qubits beyond the v3.52.0 Hopf-only tapering and reduces the Pauli term count by $\sim 3\%$ across the production hydride library, verified bit-exact across LiH (+3), HF (+6), BeH_2 (+5), H_2O

(+7), NH_3 (+8), CH_4 (+9). The relativistic jj-coupled analog has no such parity protection (the half-integer-j sum is not forced even by jj-coupled selection rules), structurally explaining why the relativistic Tier 2 chemistry construction admits no direct m_j-parity $\mathbb{Z}_2$ tapering analog. Production: `geovac/extended_tapering.py` (v3.89.0, `tapered='extended'`); detailed analysis and panel data in Paper 14 (the ℓ-parity-tapering and Gaunt-parity-protection sections).

D. Non-central interactions

Theorems 1–3 apply to central (rotationally invariant) two-body interactions. Non-central interactions—tensor forces (S_{12}), spin-orbit coupling ($\mathbf{L} \cdot \mathbf{S}$), and velocity-dependent potentials—have modified selection rules. The tensor force, for example, couples states with $|\Delta L| = 0, 2$ and mixes partial waves, expanding the set of allowed transitions relative to the central-force case. For such interactions, the angular sparsity bound of Theorem 3 is a *lower* bound on the ERI density (i.e., the Hamiltonian may be denser than the central-force prediction). The Gaunt selection rules for the central component still apply to that component, but additional nonzero terms appear from the non-central components. A complete sparsity analysis for nuclear Hamiltonians with tensor and spin-orbit forces requires extending the angular classification to include spin-dependent Gaunt-type coefficients, which is beyond the scope of this paper.

VI. UNIVERSAL VS. COULOMB-SPECIFIC STRUCTURE

The GeoVac spectral graph theory framework exhibits two distinct kinds of structure that are sometimes conflated. Theorem 2 and its corollary allow us to draw a sharp line between them.

Universal across spherical fermion systems. The following properties depend only on the angular momentum content of the basis, not on the specific radial potential:

- Radial–angular factorization of two-body matrix elements (Theorem 1).
- Gaunt-selection-rule sparsity $D(l_{\max})$ (Theorem 3).
- $O(Q^4) \to O(Q^{2.5})$ reduction from single-center to composed block architectures (Corollary 1).
- Bit-identical zero patterns across Coulomb, harmonic, Woods–Saxon, square well, and Yukawa potentials (Table I).

These properties hold for any central two-body interaction and any radial single-particle potential that produces a complete orbital basis.

Coulomb-specific. The following properties are tied to the $SO(4)$ accidental degeneracy of the hydrogen atom and do not extend to generic central potentials:

- The Fock stereographic projection from momentum space to S^3 that maps the hydrogen Schrödinger equation to an integer eigenvalue problem on the three-sphere [9].
- The π-free graph structure in which all one-body eigenvalues, degeneracies, and coupling coefficients are integers or rationals before projection to continuous manifolds.
- The universal constant $K = -1/16$ that maps graph eigenvalues to the Rydberg spectrum.
- The hydrogen shell degeneracy sequence n^2 as $O(n^2)$-dimensional $SO(4)$ irreducible representations.

A companion paper [13] establishes that the S^3 Fock projection is rigid in the following sense: modifying the Coulomb potential $-Z/r$ to any other central form breaks the $SO(4)$ symmetry, removes the shell degeneracy, and destroys the integer spectrum. The Coulomb case is the unique central potential for which the one-body Hamiltonian admits a π-free graph Laplacian interpretation.

This partition clarifies the scope of each structural feature. The angular sparsity results of this paper apply broadly across quantum-simulation domains; the S^3 graph Laplacian machinery applies only to Coulomb systems. The two classes of result should be cited and deployed accordingly.

VII. CONCLUSION

The zero structure of two-body matrix elements in a spherical harmonic basis is determined entirely by the Wigner $3j$ symbols (Gaunt coefficients) that encode angular momentum selection rules. This structure is independent of the radial potential that defines the single-particle orbitals, independent of the specific form of the two-body interaction (provided it is rotationally invariant), and independent of the number of radial quantum numbers retained in the basis.

The explicit angular ERI densities (physical Coulomb selection rule, global M_L conservation) obtained by exact enumeration are 14.84% at $l_{\max} = 1$, 8.52% at $l_{\max} = 2$, 6.06% at $l_{\max} = 3$, 4.83% at $l_{\max} = 4$, and 3.99% at $l_{\max} = 5$. Numerical verification across five potentials (Coulomb, harmonic oscillator, Woods–Saxon, square well, Yukawa) at $l_{\max} = 2$ confirms the theorem to machine precision: the angular zero patterns are bit-identical and radial zeros contribute no additional sparsity.

The universal–specific partition is sharp. Angular sparsity and radial–angular factorization are universal across spherical fermion systems; the measured $O(Q^{2.5})$ Pauli scaling of composed block architectures is expected, but not here proven, to carry over beyond the Coulomb hydride library (Corollary 1). The S^3 Fock projection, the π-free graph Laplacian, and the integer spectrum of the unperturbed hydrogen problem are Coulomb-specific and break under any modification of the radial potential [13]. Sparsity-aware quantum simulation techniques developed for electronic structure (qubit tapering, Pauli grouping, tensor factorization) transfer directly to nuclear structure, quantum dots, and other spherical-potential systems without modification, because the angular structure of the spherical harmonic basis is a computational resource that belongs to the geometry of angular momentum, not to any particular physical potential.

VIII. SPINOR-BLOCK ANGULAR SPARSITY

The potential-independence theorem of §II stated that the ERI density $d(l_{\max})$ of the scalar Coulomb Hamiltonian on S^3 depends only on $l_{\max}$ — not on the potential $V(r)$ — because the zero/nonzero structure of the ERI tensor is fixed by the Gaunt 3j triangle rules. This section extends the theorem to the jj-coupled spinor (Dirac) basis used in the Dirac-on-S^3 Tier 2 sprint, where one-body orbitals are labeled by (n, κ, m_j) with $\kappa = -(l+1)$ for $j = l + 1/2$ and $\kappa = +l$ for $j = l - 1/2$.

A. Restatement in the spinor basis

The claim is that the spinor ERI density $d_{\text{spinor}}(l_{\max})$, defined as the fraction of nonzero entries in the 4-index Coulomb tensor over the jj-coupled one-body basis $\{|n, \kappa, m_j\rangle : l(\kappa) \le l_{\max}\}$, depends only on $l_{\max}$ and is independent of $V(r)$. The proof is structurally identical to the scalar case: the full-Gaunt jj-coupled Coulomb reduction (Dyall–Faegri [15], Grant [16]),

$$\langle ab\,|\,V\,|\,cd\rangle \neq 0 \iff \exists k \ge 0 \text{ satisfying}$$

1. parity: $(l_a + l_c + k)$ and $(l_b + l_d + k)$ both even,
2. j-triangles: $|j_a - j_c| \le k \le j_a + j_c$ and analogous for (b, d),
3. reduced 3j: $\begin{pmatrix} j_a & k & j_c \\ 1/2 & 0 & -1/2 \end{pmatrix} \neq 0$ and analogous for (b, d),
4. bottom-row 3j: $\begin{pmatrix} j_a & k & j_c \\ -m_{j,a} & m_{j,a}-m_{j,c} & m_{j,c} \end{pmatrix} \neq 0$ and analogous for (b, d),
5. m-conservation: $m_{j,a} + m_{j,b} = m_{j,c} + m_{j,d}$,

gives a selection rule whose triangle and parity conditions depend only on $(l_a, l_b, l_c, l_d, j_a, j_b, j_c, j_d, m_{j,a}, m_{j,b}, m_{j,c}, m_{j,d}, k)$ — not on any radial matrix element R^k or any property of $V(r)$. Replacing Gaunt 3j by the pair of jj-coupled reduced 3j symbols (one per pair) preserves the V-independence; the radial factor $R^k(n_a l_a, n_b l_b, n_c l_c, n_d l_d)$ appears as a multiplicative weight on each nonzero entry but does not change the zero/nonzero pattern.

B. Numerical values of $d_{\text{spinor}}(l_{\max})$

Table II reports $d_{\text{spinor}}(l_{\max})$ for $l_{\max} = 0, \ldots, 5$, in both the *pair-diagonal-m* convention that Paper 22's existing scalar table uses (stricter rule: $m_a = m_c$ and $m_b = m_d$ per pair), and the *full-Gaunt* convention (physically correct Coulomb selection rule: $m_{j,a} + m_{j,b} = m_{j,c} + m_{j,d}$ global). The two conventions give quantitatively similar ratios; both scalar conventions now appear in Paper 22 §II (D = full-Gaunt physical density, D_{pd} = pair-diagonal), and the $d_{\text{sc}}^{\text{FG}}$ column here reproduces the headline $D(l_{\max})$ bit-exactly. All densities are exact rationals from `sympy.Fraction`; percentages are their 4-decimal renderings.

TABLE II. Spinor-block angular sparsity density $d_{\text{spinor}}(l_{\max})$ in the jj-coupled basis, compared to the scalar density $d_{\text{scalar}}(l_{\max})$ of §II. $Q_{\text{scalar}} = (l_{\max}+1)^2$, $Q_{\text{spinor}} = 2(l_{\max}+1)^2$. Both conventions are tabulated in Paper 22 §II: the full-Gaunt convention is the physically correct Coulomb selection rule (the headline $D(l_{\max})$), and the pair-diagonal-m convention is the stricter axial sub-case (D_{pd}).

	pair-diagonal-m convention						full-Gaunt conv	
$l_{\max}$	Q_{sc}	Q_{sp}	d_{sc}	d_{sp}	ratio (exact)	ratio	$d_{\text{sc}}^{\text{FG}}$	d
0	1	2	100.0000%	25.0000%	1/4	0.2500	100.0000%	25.0
1	4	8	7.8125%	4.2969%	11/20	0.5500	14.8438%	8.59
2	9	18	2.7587%	2.1071%	553/724	0.7638	8.5200%	6.45
3	16	32	1.4404%	1.2329%	101/118	0.8559	6.0608%	5.16
4	25	50	0.9024%	0.8106%	2533/2820	0.8982	4.8274%	4.30
5	36	72	0.6190%	0.5722%	9611/10396	0.9245	3.9863%	3.67

C. Structural interpretation

Three observations on Table II.

(i) $d_{\text{spinor}} \le d_{\text{scalar}}$ at every $l_{\max}$, monotonically. The j-triangle rules can only add zeros on top of the l-triangle and parity rules; they never remove a forbidden transition. The spinor density is therefore bounded above by the scalar density in both conventions.

(ii) The ratio $d_{\text{spinor}}/d_{\text{scalar}}$ saturates to 1 as $l_{\max}$ grows. From 0.25 at $l_{\max} = 0$ (pure spin-dilution floor, single $s_{1/2}$ orbital with two m_j branches) to 0.92 at $l_{\max} = 5$. The structural reason: the j-triangle $|j_a - j_c| \le k \le j_a + j_c$ differs from the l-triangle $|l_a - l_c| \le k \le l_a + l_c$ by at most 1 (recall $j = l \pm 1/2$). The extra zeros are boundary corrections to the triangle set, and their relative size shrinks as $O(1/l_{\max})$ as $l_{\max}$ grows.

(iii) Prefactor inflation at matched $l_{\max}$. Absolute Pauli count in the spinor basis is inflated by $(2Q_{\rm scalar}/Q_{\rm scalar})^4 = 16\times$ from the quadrupling of index ranges; times the density ratio $d_{\rm spinor}/d_{\rm scalar} \approx 0.86$ at $l_{\max} = 3$, the absolute nonzero count is $\sim 14\times$ larger. At matched *qubit count* Q — the regime relevant to Paper 14's spinor-composed resource table — the multiplier is milder: $\sim 2.4\times$ at $n_{\max} = 2$, $\sim 5.9\times$ at $n_{\max} = 3$. The distinction between matched-$l_{\max}$ and matched-Q multipliers is structurally important and should be stated alongside any sparsity-exponent comparison.

D. Sharpened universal/Coulomb-specific partition

Paper 22 §VI identified angular sparsity as a *universal* property of the S^3 angular basis, in contrast to the S^3 Fock projection and the Hopf bundle which are *Coulomb-specific* structures. Tier 2's verification tightens this statement:

> Angular sparsity is universal across scalar Schrödinger *and* spinor Dirac systems. The sparsity density depends only on the multiplet structure of the angular basis (Gaunt 3j for scalars, jj-coupled reduced 3j for spinors), not on the radial Hamiltonian or its spectrum. The S^3 Fock projection and the BJL algebra that would otherwise lift it to the Dirac sector remain scalar-Schrödinger- specific structures — the spinor basis can be built on the same S^3 graph, but the exact mapping of hydrogenic shells to graph nodes is a scalar phenomenon (Tier 1 Explorer Finding 2.1).

The Tier 2 extension therefore *strengthens* Paper 22's universal result — the same potential-independence theorem now applies to scalar and spinor Hamiltonians on the same S^3 basis — while preserving the Tier-1 finding that the Fock map itself does not lift.

E. Implementation and verification

Both spinor columns of Table II are pinned to exact nonzero counts and 4-decimal percentages by the regression test `tests/test_paper22_spinor_density.py`, which re-enumerates the jj-coupled selection rule in exact `sympy` arithmetic over the full single shell ($l_{\max} = 0, \ldots, 5$; the four heaviest tuples are `@slow`-marked). All selection-rule decisions are sympy symbolic (`wigner_3j(...) == 0` is a symbolic boolean); no floating point enters the zero/nonzero decision. The discovery-mode driver (Dirac-on-S^3 Tier 2 Track T0) is archived at `debug/archive/spectral_triple_arc/tier2_t0_spinor_density.py`, and its data dump — integer numerators and denominators for every density — remains at `debug/data/tier2_t0_spinor_density.json`.

The full-Gaunt scalar column $d^{\rm FG}_{\rm sc}$ of Table II reproduces the headline density $D(l_{\max})$ pinned by `tests/test_paper22_density.py` bit-exactly, so the scalar/spinor boundary is mutually consistent across the two tests and the selection-rule implementation is validated against the existing reference.

IX. RANK-2 EXTENSION: BREIT INTERACTION

Sprint 2 Track BR-A (April 2026) extends the angular sparsity theorem from the rank-0 Coulomb operator $1/r_{12}$ to the rank-2 Breit-Pauli spin-spin tensor

$$H_{SS} = \alpha^2 \frac{[\sigma_1 \otimes \sigma_2]^{(2)} \cdot Y^{(2)}(\hat{r}_{12})}{r_{12}^3}$$

and the combined rank-1 + rank-2 spin-other-orbit operator $H_{SOO} \propto \alpha^2(\ell_1 \cdot \sigma_2 + \ell_2 \cdot \sigma_1)/r_{12}^3$, both in the jj-coupled spinor basis of §VIII. The resulting angular density $d_{\rm Breit}(l_{\max})$ for the union of non-zero matrix elements in $\{H_{SS}, H_{SOO}\}$ at $Z = 4$:

$l_{\max}$	Q	$d_{\rm Coulomb}$	d_{SS}	d_{SOO}	$d_{\rm Breit}$	$d_{\rm Breit}/d_{\rm Coulomb}$
0	2	25.00%	0.00%	0.00%	25.00%	1.00×
1	8	8.59%	31.25%	22.27%	53.91%	6.27×
2	18	6.46%	28.88%	18.99%	47.89%	7.41×
3	32	5.17%	24.73%	15.57%	40.30%	7.80×
4	50	4.30%	21.12%	13.03%	34.15%	7.94×
5	72	3.68%	18.25%	11.15%	29.40%	7.99×

Numerical result: the angular sparsity bound extends to rank-2 (verified at $Z = 4$, $l_{\max} \le 5$; formal proof deferred). The Breit density is 6–8× denser than Coulomb (consistent with rank-2 being less restrictive than rank-0 in the triangle-inequality sense: $|\Delta l| \le 2, |\Delta j| \le 2, |\Delta m_j| \le 2$ vs $= 0$ for Coulomb), but it remains *monotonically decreasing* with $l_{\max}$ and is bounded away from 1 as $l_{\max} \to \infty$. The saturation at $\sim 8\times$ reflects the asymptotic selection-rule density of rank-2 vs rank-0 Gaunt structure on the spinor manifold.

The proposed generalized statement (numerically supported across the cases below; formal proof deferred): *ERI density in the S^3 spinor basis depends only on $l_{\max}$ and the tensor rank k of the interaction operator, not on the radial kernel $V(r)$.* The rank-0 Coulomb (§II), rank-0 spinor (§VIII), rank-1 dipole (implicit in Tier 2 SOO), and rank-2 Breit cases all verify this form. A proof by Wigner-Eckart reduction on the jj-coupled basis is deferred to a formal write-up.

Caveat on the radial kernel. The $1/r_{12}^3$ kernel is *distributional* — its bare partial-wave expansion has a logarithmic divergence at $r_1 = r_2$, so matrix elements are not defined as ordinary integrals. The Breit-Pauli tensor projection (transverse component of $\sigma_1 \otimes \sigma_2$) regularizes this into a smooth kernel $K_l^{BP} = r_<^l / r_>^{l+3}$ with exact rational matrix elements for same-n orbitals and log-transcendental content for cross-n (see Paper 18 for the taxonomy classification). Paper 22's angular sparsity statement concerns the zero/nonzero *pattern* of the matrix, which is determined by the angular selection rules and is insensitive to the radial-kernel regularization—this is precisely why the theorem is potential-independent.

The $Z = 4$ rank-0/rank-1/rank-2 density table above (rows $l_{\max} \leq 3$) is regression-pinned in `tests/test_paper22_breit_angular.py`, which recomputes the Coulomb, spin-other-orbit, and spin-spin angular densities from the Wigner-3j selection rules and checks the potential-independence (kernel-truncation invariance) of the rank-0 pattern.

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